

Understanding ML, DL, NLP course

①

sec - 3.5

Adaboost ML
Algorithm

lec - 4 Updating weights

lec - 5 Normalising
weights & assigning
Bins

lec - 6 Selecting New Weak
Points

lec - 7 Final prediction of
Adaboost.

=> Update the weights of
correctly & incorrectly
classified points

Salary	Credit	Approval	sample weight
<= 50K	B	No	1/7
<= 50K	G	yes	1/7
<= 50K	G	yes	1/7
> 50K	B	No	1/7
> 50K	G	yes	1/7
> 50K	N	yes	1/7
<= 50K	N	No	1/7

Performance of Stuck

$$= \frac{1}{2} \ln \left[\frac{1 - TE}{TE} \right]$$

$$= \frac{1}{2} \ln \left[\frac{1 - \frac{1}{2}}{\frac{1}{2}} \right]$$

$$= \frac{1}{2} \ln 1 \approx 0.89$$

$$\phi = \alpha_1 (m_1) + \alpha_2 (m_2) + \dots$$

$$\phi = 0.896$$

Salary	Credit	Approval	sample (3) weights
<= 50K	B	No	1/7
<= 50K	G	Yes	1/7
<= 50K	G	Yes	1/7
> 50K	B	No	1/7
> 50K	G	Yes	1/7
> 50K	N	Yes	1/7
<= 50K	N	No	1/7

Update
wts

For correct classified
samples

$$= \text{weight} * e^{-\text{Information of Sample}}$$

$$= \frac{1}{7} * e^{-(0.894)}$$

$$= 0.088$$

(4)
 $\Rightarrow$ For Incorrect classified
 misclassified

$$= \text{weight} * C_{\text{misclassified}}$$

$$= \frac{1}{7} * C_{\text{misclassified}} \\ = 0.349$$

$\Rightarrow$ For correct classified
 correct

$$= \text{weight} * C_{\text{correct}}$$

Salary	Credit	Approval	sample weights
$< 50K$	B		
$< 50K$	G		
$< 50K$	G		
$> 50K$	B		
$> 50K$	G		
$> 50K$	G		
$< 50K$			

2) Normalizing weights > Assigning Bins

5

Sales	Credit	Approval	Weight
L = 50K	B	No	0.058
L = 50K	G	Yes	0.058
L = 50K	G	Yes	0.058
L = 50K	B	No	0.058
> 50K	G	Yes	0.058
> 50K	G	Yes	0.058
> 50K	N	Yes	0.058
L = 50K	N	No	0.058
			0.001

Normalized weights

0.08
 0.08
 0.08
 0.08
 0.08
 0.50
 0.08

 2.1

=> Bins Assignments

⑥

0 - 0.08

0.08 - 0.16

0.16 - 0.24

0.24 - 0.32

0.32 - 0.40

0.40 - 0.50

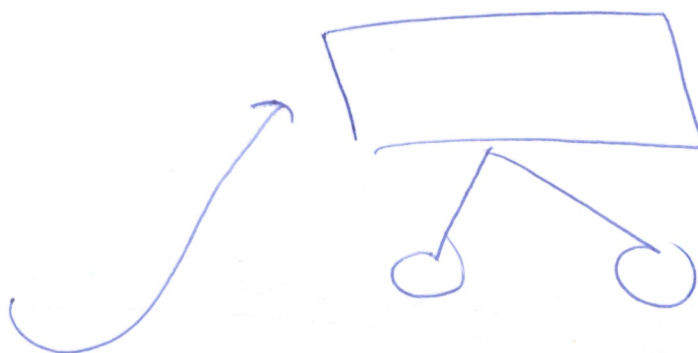
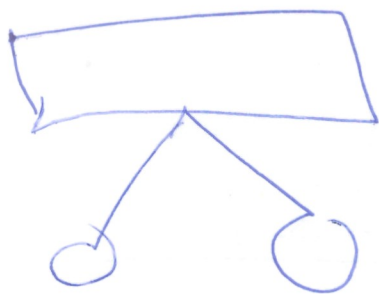
0.50 - 0.58

performance of street ≈ 0.896

$$Q = d_1(m_1) + d_2(m_2) + \dots + d_n(m_n)$$

$$d_1 \approx 0.896$$

performance



④ select the data point
thrusts test out

① Initiate the process
selecting random value α/w
0.5

S	Credit	Approval	Random
>50K	N	yes	0.50
<=50K	G	yes	0.10
>50K	N	yes	0.60
>50K	N	yes	0.75
<=50K	G	yes	0.24
>50K	B	no	0.32
>50K	N	yes	0.81

performance strength $\Rightarrow 0.65$

$$y_i = x_1(x_1) + x_2(x_2)$$

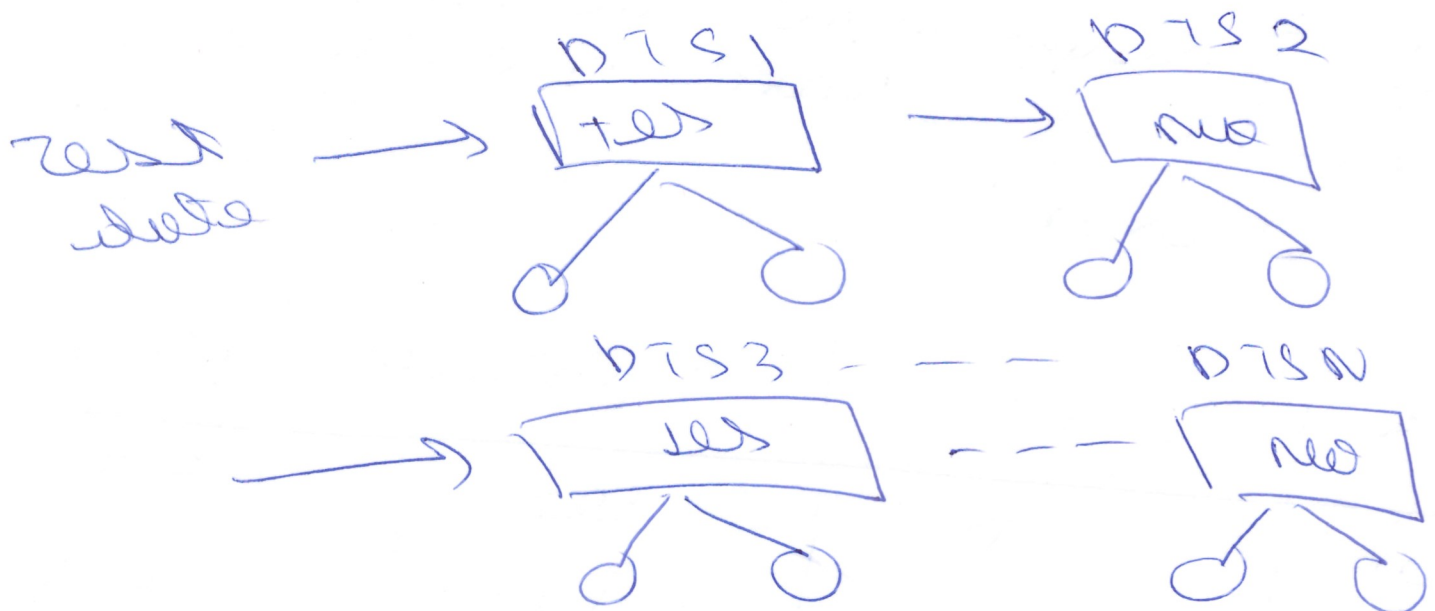
$$x_1 = 0$$

$$x_2 = 0.05$$

④ Find Prediction

⑧

test data ($\leq 50\%$, 0)



$$\alpha_1 = 0.816$$

$$\alpha_3 = 0.24$$

$$\alpha_2 = 0.650$$

$$\alpha_N = -0.30$$

$$\theta = \alpha_1(m_1) + \alpha_2(m_2) + \alpha_3(m_3) + \dots + \alpha_N(m_N)$$

$$= 0.816(\text{yes}) + (0.650)(\text{no}) + 0.24(\text{yes}) - 0.30(\text{no})$$

$$= 1.136(\text{yes}) + 0.350(\text{no})$$

→ performance of say (yes) ≥ 1.136

performance of say (no) ≥ 0.350